

**LESSON 138 (0.5 GROUND and 1.5 SOLO CROSS COUNTRY)****Lesson Objective**

The student will use the previously learned cross-country skills on a solo cross-country flight. This flight will increase the student's proficiency and confidence. See supplemental handouts for recommended routes and airports.

**Briefing**

- Review homework assigned in the previous lesson.
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

**Lesson Content (Ground)****Instructor Review of Flight Plan**

- Review the entirety of the nav log and flight plan.
  - Make sure there are no errors, and everything is filled out correctly.
- Review the weather briefing obtained by the student.
- Endorse the student.

**Review Cross-Country Flight Planning**

- Airspace
  - Review the student's chosen route and verify the student's consideration regarding airspace.
  - Cruising altitudes should be appropriate for the airspace.
- Weight and Balance
  - Review the student's weight and balance planning on the TOLD card.
  - Verify that the fuel information on the TOLD card matches the fuel information on the navlog.
- Performance and Limitations
  - Review the student's calculations for pressure altitude and density altitude.
  - Verify that the V speeds were filled out correctly on the TOLD card.
  - Inspect the totals (times, fuel, distances).
- Navigation Log
  - Verify the entirety of the nav log.
  - Cruising altitudes should be appropriate for the magnetic course.
  - Ensure that the airport diagrams are drawn, and the frequencies are filled in.
  - Verify that the student has the line drawn and checkpoints identified in the sectional chart.
  - Ensure that the student has a TOC and TOD checkpoint.
  - The information on the fuel management table must match the information on the front of the nav log.

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- Verify that the flight plan portion has the required and correct entries.
- VOR information should be written.
- Weather
  - Review the student's weather planning.
  - Ensure that the student obtained a standard weather briefing.
  - Ask the student questions about any potential weather hazards for the route of flight.
  - Ensure that the student has filled out the weather section of the navlog.
  - Cruising altitudes should be appropriate for the weather.
- Flight Plan Filed with FSS
  - Ensure that the student has filed the flight plan with FSS.
  - Verify departure, enroute, and arrival times are correct.
- Brief the student:
  - Opening the flight plan.
  - Getting flight following.

**Go/no-go decision**

- The student must make a go/no-go decision depending on:
  - Personal minimums.
  - FIT minimums.
  - Aircraft.
  - Airspace.
- Review the students planning with a focus on:
  - Weather, weight and balance, performance, flight planning, and fuel calculations.
  - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
  - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
  - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

**Lesson Content (Flight)****Cross-Country Flight Planning**

- Student pilot solo.

**Weight and Balance**

- Student pilot solo.

**Fuel requirements**

- Student pilot solo.

**Charts and Publications**

- Student pilot solo.

**Performance and Limitations**

- Student pilot solo.

**Flight Plan (filed, opened & closed)**

- Student pilot solo.

**Navigation Systems and Radar Services**

- Student pilot solo.

**Pilotage**

- Student pilot solo.

**Dead Reckoning**

- Student pilot solo.

**Traffic Patterns**

- Student pilot solo.

**Ground Speed Calculations**

- Student pilot solo.

**Estimating ETA's**

- Student pilot solo.

**Normal Takeoffs**

- Student pilot solo.

**Normal Landings**

- Student pilot solo.

**Postflight Procedures**

- Student pilot solo.

**Post-Brief**

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 139.
- Assign homework.

**Completion Standards**

The student shall adequately plan and safely conduct solo cross-country operations using the three methods of navigation.